Homework 3 **Vishal Kevat**

HW3 Virtuoso Repo: https://github.com/vishalkevat007/Advanced-VLSI-Design-ECE695-Spring2026-Homeworks.git

(a) Implement a clock gater based on standard transmission gate latch, simulate and verify the clock gating functionality (both clock gating and ungating) and obtain the clock energy from your simulations.

Solution:

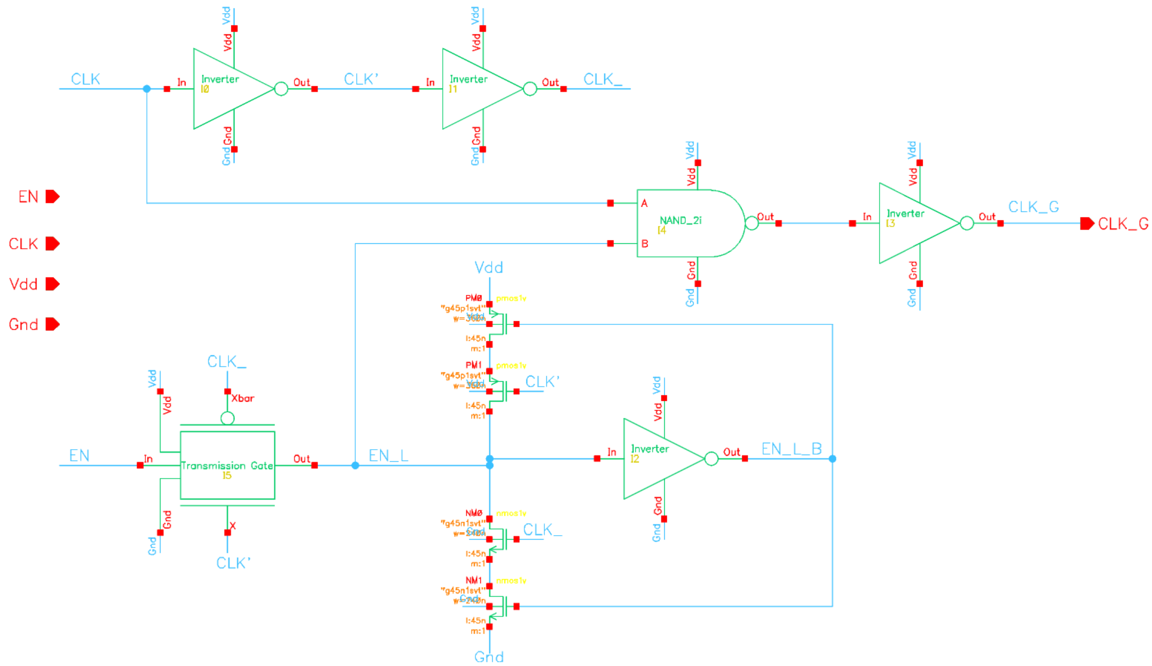

Fig 1: Clock Gater based on Standard Transmission Gate Latch

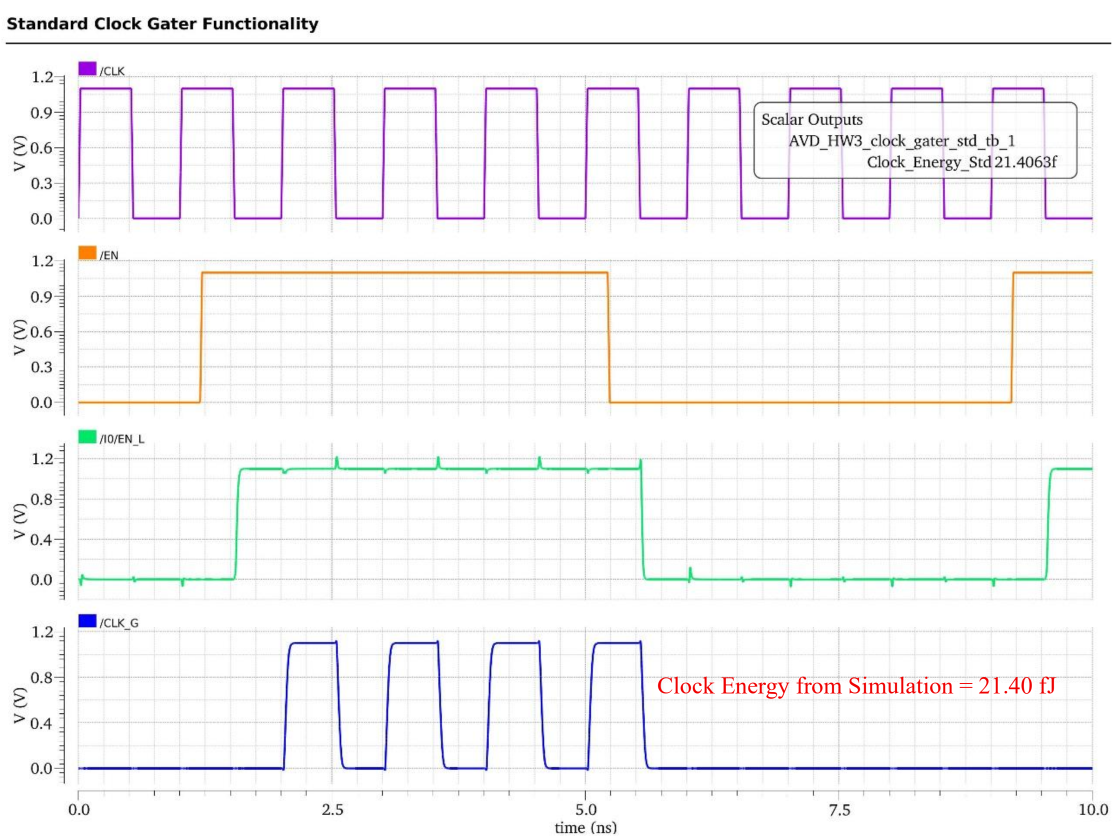

Fig 2: Clock Gating (Transmission Gate based) functionality - Gating when EN=0, Ungating when EN=1

Clock Energy Calculation

A load capacitance of **2 fF** is connected at the ***CLK_G* output** in the testbench to model the load driven by the clock gater during switching.

The clock energy is obtained by integrating the instantaneous power drawn from the supply during the interval when the clock is ungated.

$$P(t) = V_{DD}(t)\, I_{DD}(t)$$

$Where,\ \ V_{DD}(t) = 1.1$ V (DC supply) ; $\ I_{DD}(t) = Instantaneous$ current through the V_{DD} $\{IT("/Vdd")\}$

The total clock energy is:

$$E_{clock} = \int_{t_1}^{t_2} V_{DD}\,(t)\, I_{DD}(t)\, dt$$

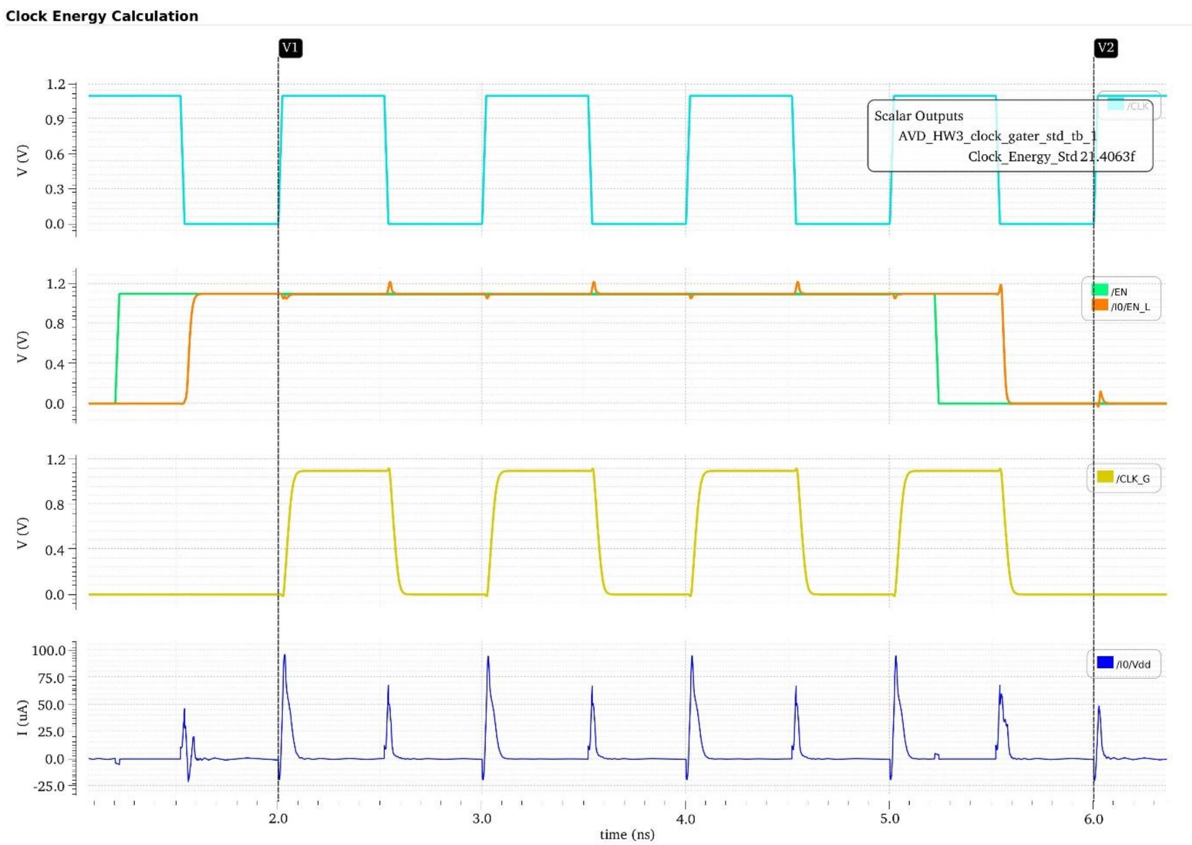

Fig 3: Clock Energy Calculation window

The integration window corresponds to the interval where **EN = 1 and CLK_G toggles**.

$$E_{clock} = \int_{2\,ns}^{6\,ns} V_{DD}\,(t)\, I_{DD}(t)\, dt$$

Using the Cadence calculator expression: *integ(IT("/Vdd") * VT("/vdd!") 2n 6n)*; the total clock energy obtained from simulation is:

$$E_{clock} = \mathbf{21.40\ fJ}$$

This value represents the total energy drawn from VDD over the ungated interval from 2 ns to 6 ns
Similarly, the Clock energy was calculated for the SR latch based Clock Gater.

(b) Repeat the same for the race-free implementation based on set-reset latch with two transistors in stack.

Solution:

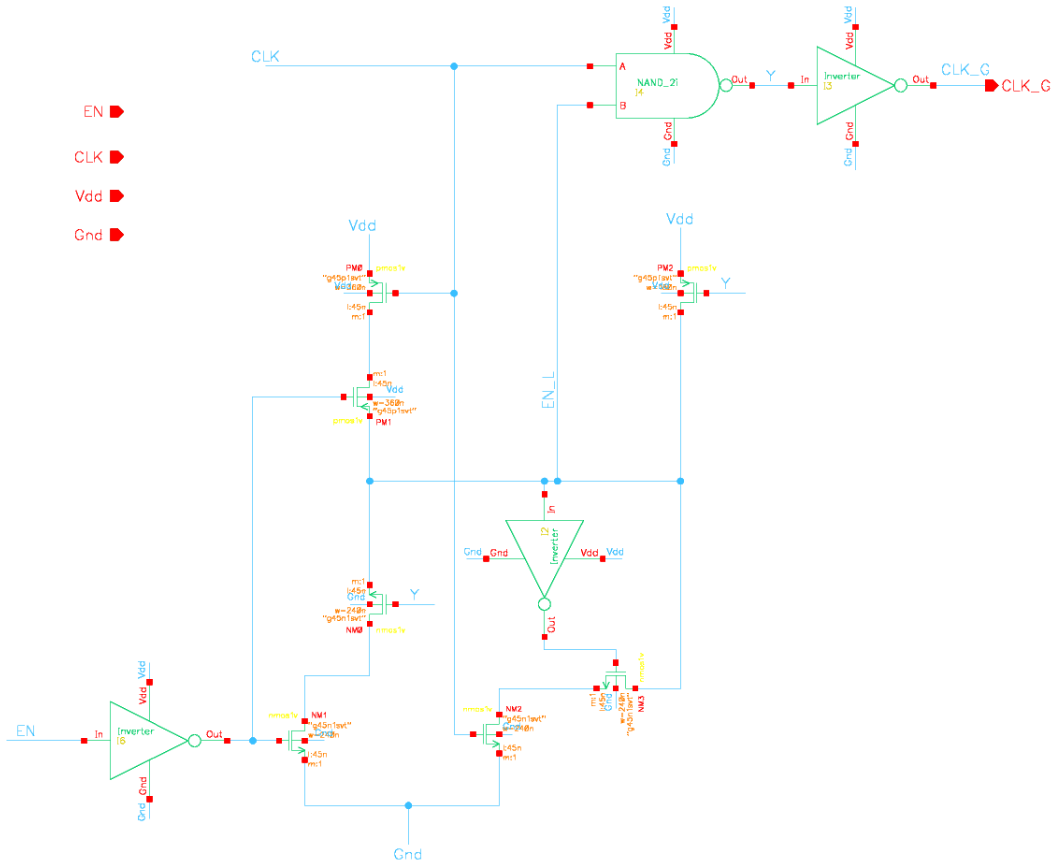

Fig 4: Clock Gater based on Set-Reset latch

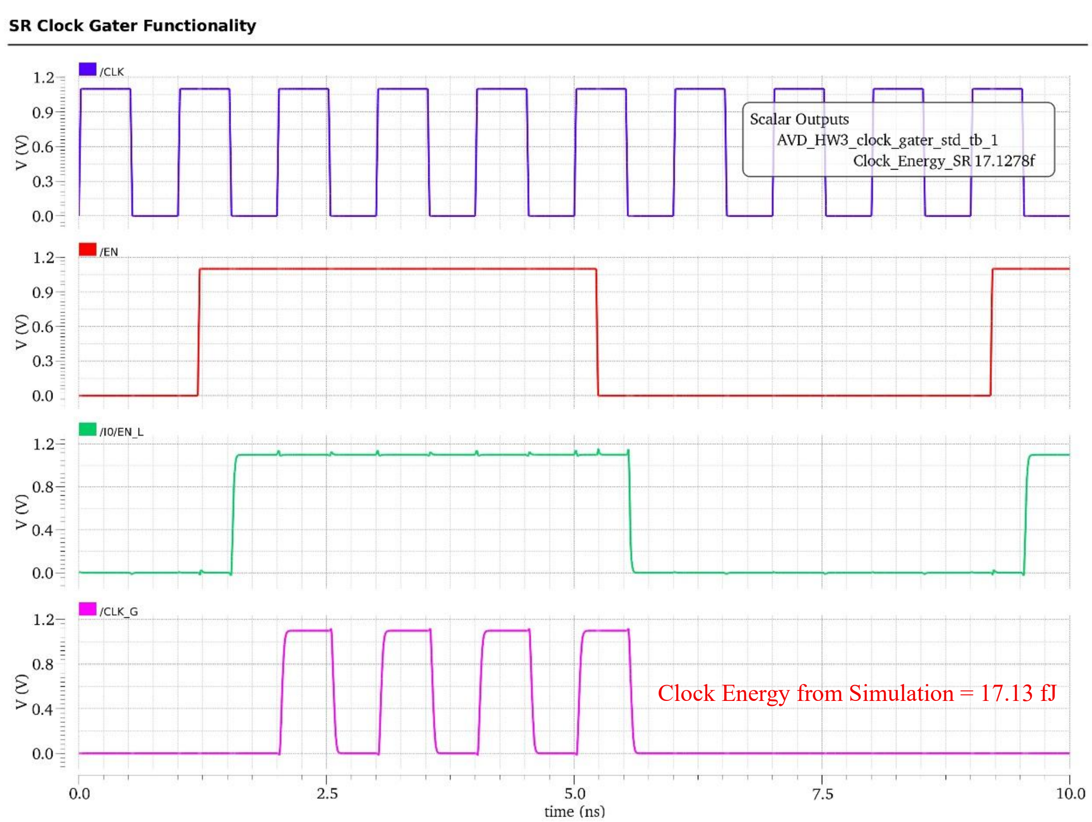

Fig 5: Clock Gating (SR based) functionality - Gating when EN=0, Ungating when EN=1

(c) Compare the clock energies for the two cases.

Solution:

Clock Gater Designs	**Energy During Ungating EN=1 (fJ)**	**Comments**
Standard Transmission Gate Latch based	21.40	Higher clock energy because the clock drives more internal nodes and transistors (through CLK_, CLK′ and the transmission gate latch). This increases the effective clock capacitance and therefore increases the dynamic energy drawn from the supply.
Set-Reset Latch based (Race free with two stacked transistors)	17.13	Lower clock energy because fewer transistors and internal nodes are directly driven by the clock. The reduced clock loading results in lower switching activity and therefore lower energy consumption during the ungated interval.

The SR latch based clock gater implementation reduces total clock energy by approximately **20%** relative to the transmission gate latch based clock gater under identical testbench conditions.